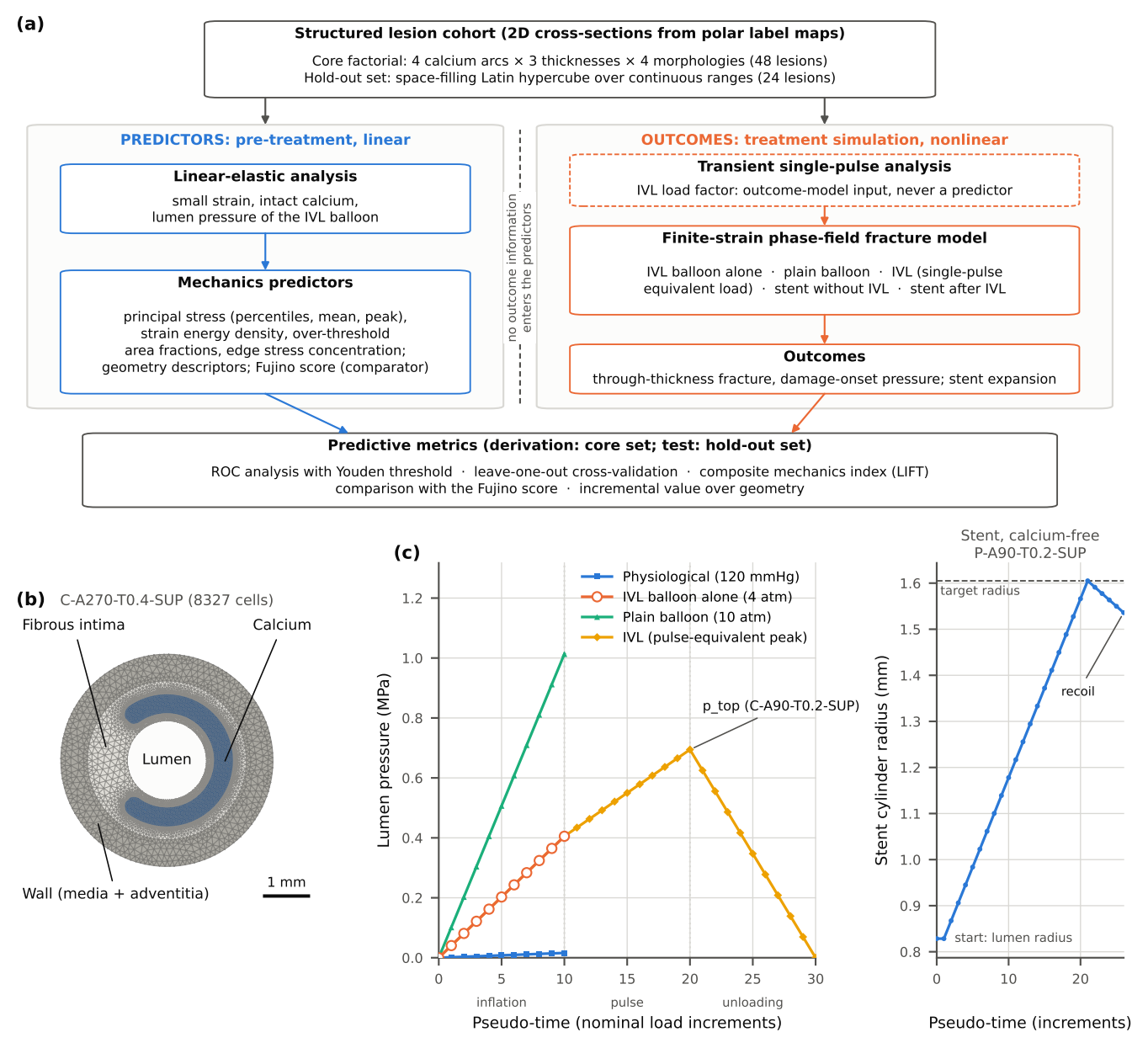

(a)
Structured lesion cohort (2D cross-sections from polar label maps)
Core factorial: 4 calcium arcs × 3 thicknesses × 4 morphologies (48 lesions)
Hold-out set: space-filling Latin hypercube over continuous ranges (24 lesions)
PREDICTORS: pre-treatment, linear
Linear-elastic analysis
small strain, intact calcium,
lumen pressure of the IVL balloon
Mechanics predictors
principal stress (percentiles, mean, peak),
strain energy density, over-threshold
area fractions, edge stress concentration;
geometry descriptors; Fujino score (comparator)
no outcome information enters the predictors
OUTCOMES: treatment simulation, nonlinear
Transient single-pulse analysis
IVL load factor: outcome-model input, never a predictor
Finite-strain phase-field fracture model
IVL balloon alone · plain balloon · IVL (single-pulse equivalent load) · stent without IVL · stent after IVL
Outcomes
through-thickness fracture, damage-onset pressure; stent expansion
Predictive metrics (derivation: core set; test: hold-out set)
ROC analysis with Youden threshold · leave-one-out cross-validation · composite mechanics index (LIFT)
comparison with the Fujino score · incremental value over geometry
(b)
C-A270-T0.4-SUP (8327 cells)
Fibrous intima
Calcium
Lumen
Wall (media + adventitia)
1 mm
(c)
Physiological (120 mmHg)
IVL balloon alone (4 atm)
Plain balloon (10 atm)
IVL (pulse-equivalent peak)
p_top (C-A90-T0.2-SUP)
Lumen pressure (MPa)
inflation
pulse
unloading
Pseudo-time (nominal load increments)
Stent, calcium-free
P-A90-T0.2-SUP
target radius
recoil
start: lumen radius
Stent cylinder radius (mm)
Pseudo-time (increments)